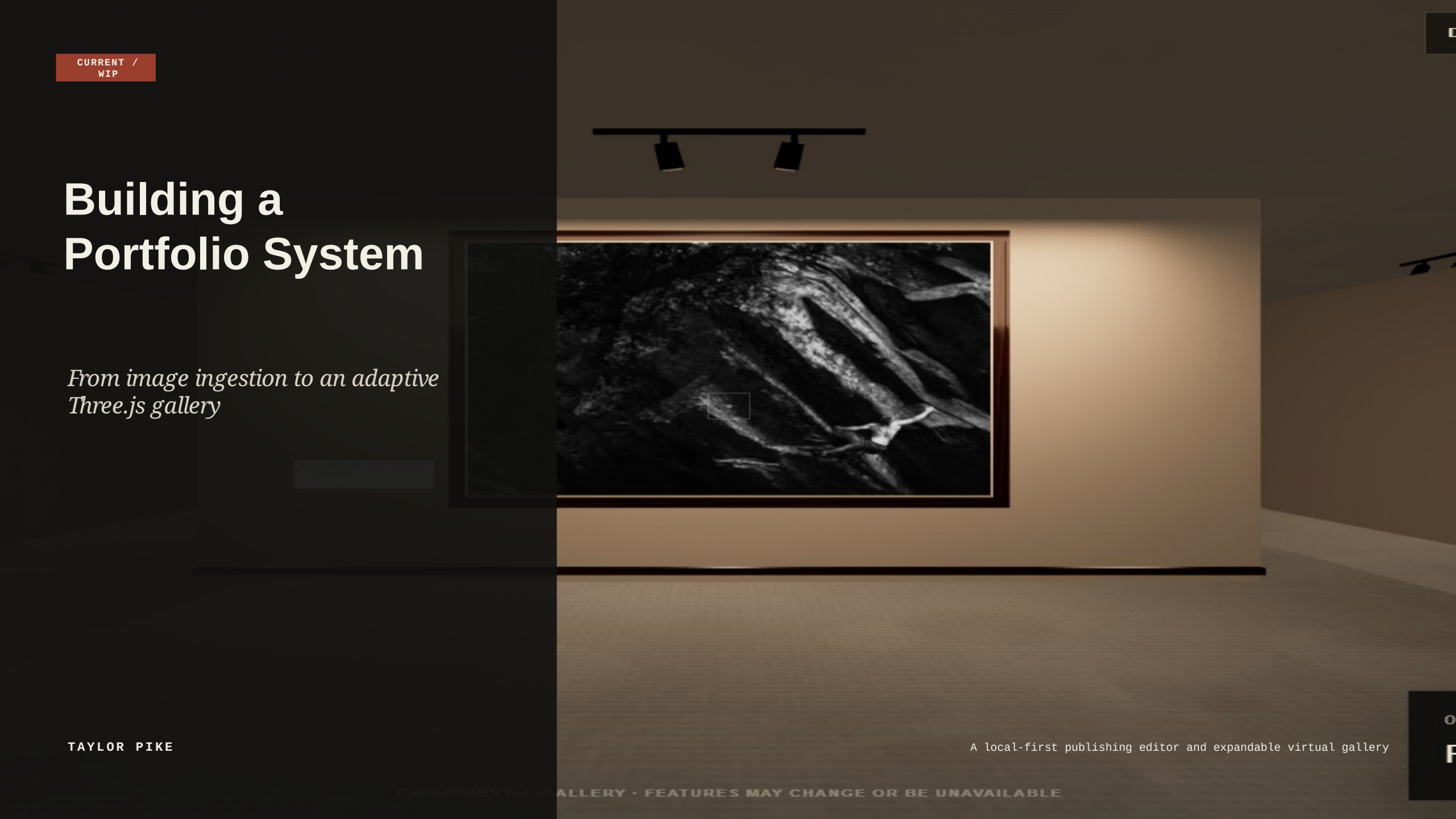
CURRENT / WIP

Building a Portfolio System

From image ingestion to an adaptive Three.js gallery



TAYLOR PIKE

A local-first publishing editor and expandable virtual gallery

EXPERIMENTAL GALLERY · FEATURES MAY CHANGE OR BE UNAVAILABLE

01 / ORIGIN

CURRENT

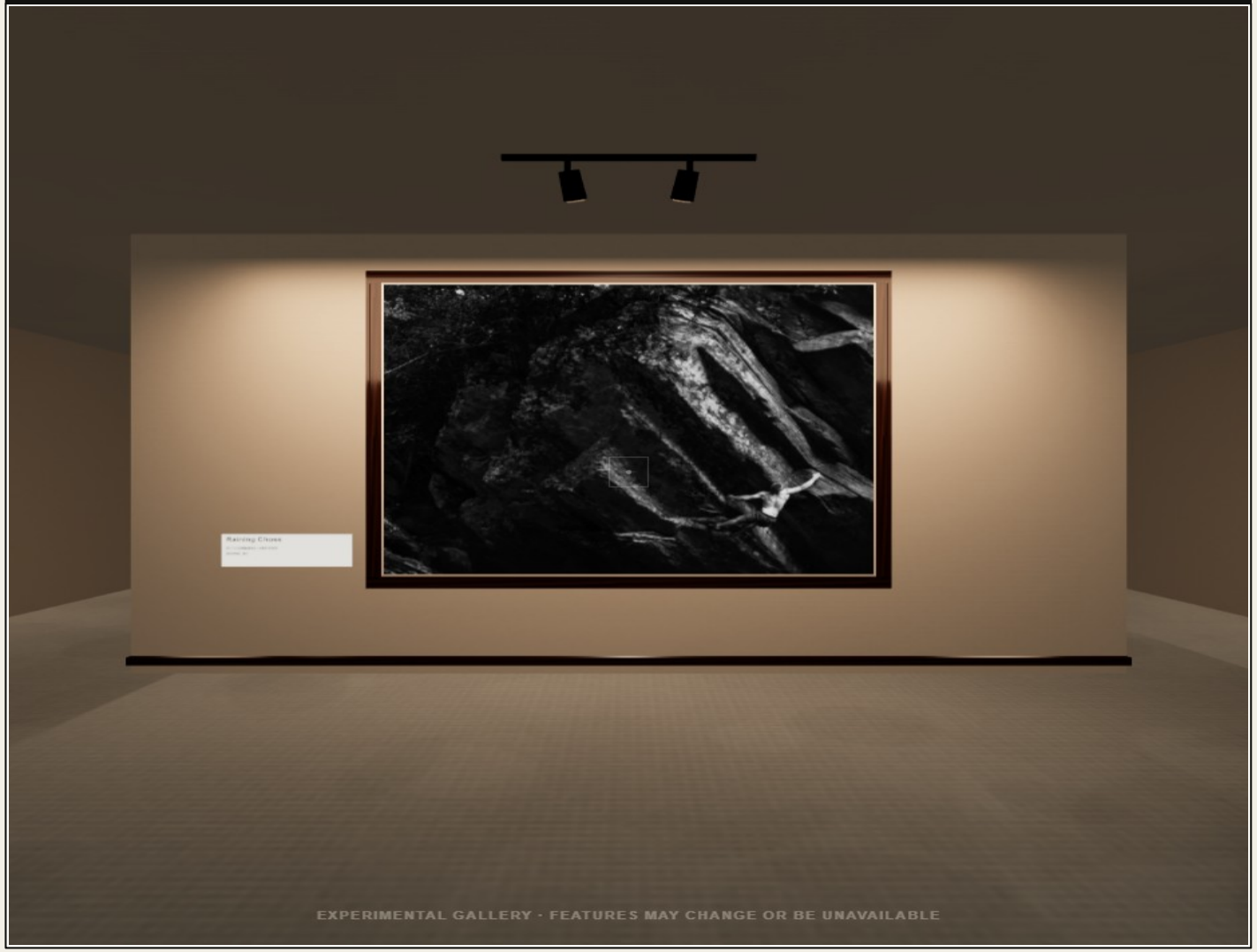
Why I built it

I wanted my photography portfolio to feel distinct from a conventional image grid or template-based site. After learning Three.js, I began thinking about a virtual gallery where visitors could move through the work rather than only scroll past it.

The gallery also continued an older ambition: since learning to code in high school, I had wanted to build an experimental JavaScript project that connected directly to my creative practice.

> *“I wanted the portfolio to feel uniquely mine and meaningful enough to leave a lasting impression.”*

VIRTUAL GALLERY / CONCEPT ORIGIN



A spatial presentation built around movement, scale, and sequence.

02 / EVOLUTION

CURRENT

How the gallery became an editor

What started as a convenience became the operating layer for the portfolio.

The first problem was managing the photographs that would eventually appear inside the gallery. Renaming files, creating multiple sizes, editing paths, and changing JSON by hand would become unmanageable as the library grew.

I began with an import tool and controls for the homepage slideshow and traditional portfolio. Each new part of the website introduced another decision that was better represented through a dedicated interface than direct source-code editing.

01 **Virtual gallery idea**

02 **Image import tool**

03 **Homepage + portfolio curation**

04 **Metadata + crop controls**

05 **About-page composition**

06 **Gallery architecture + wall curation**

07 **Site-wide control panel**

PROJECT LOGIC

Each interface emerged from a real maintenance problem inside the portfolio.

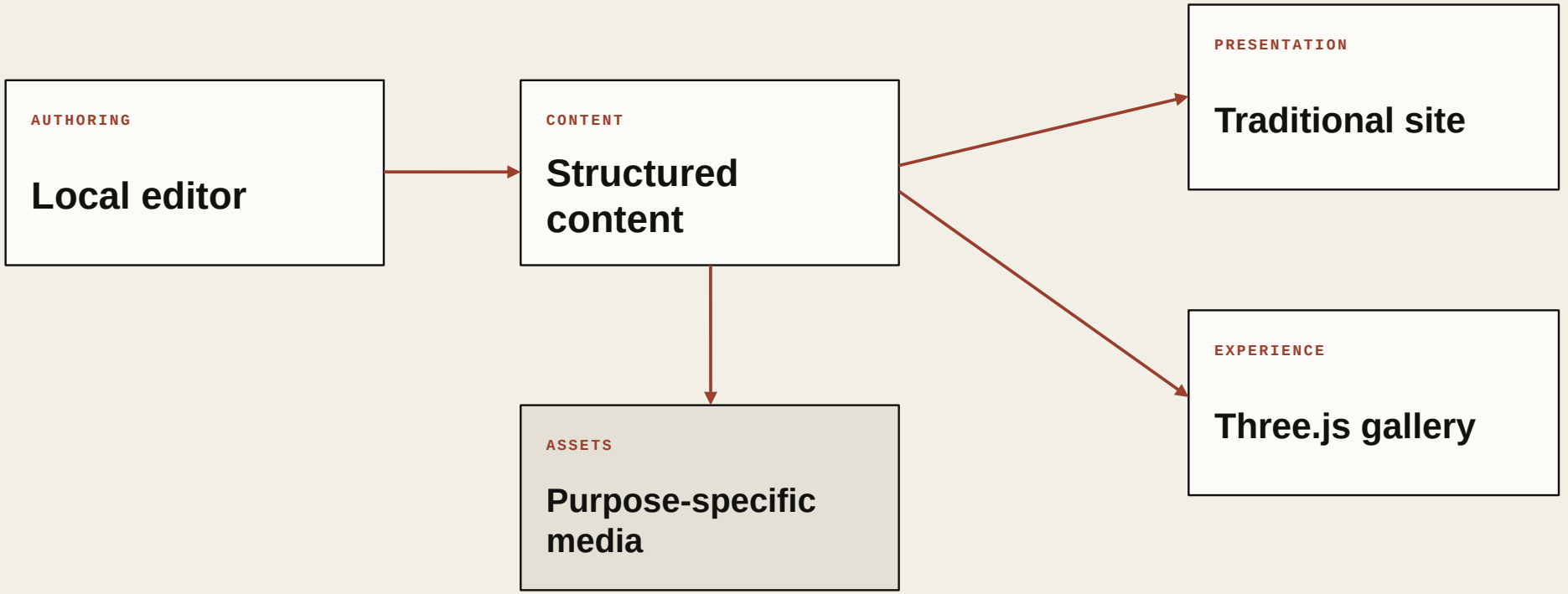
The editor was not designed as a generic CMS first and applied afterward.

03 / SYSTEM

CURRENT

One system, several responsibilities

The portfolio is a connected system: a local editor, structured content, a media pipeline, data-backed site copy, a traditional public site, and a real-time gallery. Clear boundaries let each layer evolve without forcing routine content changes into application code.



SYSTEM RULE

Content can change without rewriting the site.

Specialized interfaces appear only where a conventional form cannot represent the decision clearly.

One content layer supports multiple forms of presentation.

04 / INGESTION

CURRENT

One photograph, four responsibilities

The importer turns a source image into predictable assets and a structured record.

A thumbnail, a gallery texture, a standard display image, and a high-resolution image have different performance and quality requirements. The pipeline creates each rendition for a specific use instead of forcing every page and device to download the same oversized asset.

FAST	BALANCED	BOUNDED	DETAILED
library + grid	public display	Three.js texture	close viewing

The processed original moves to an imported-source archive after a successful import.

05 / CONTENT

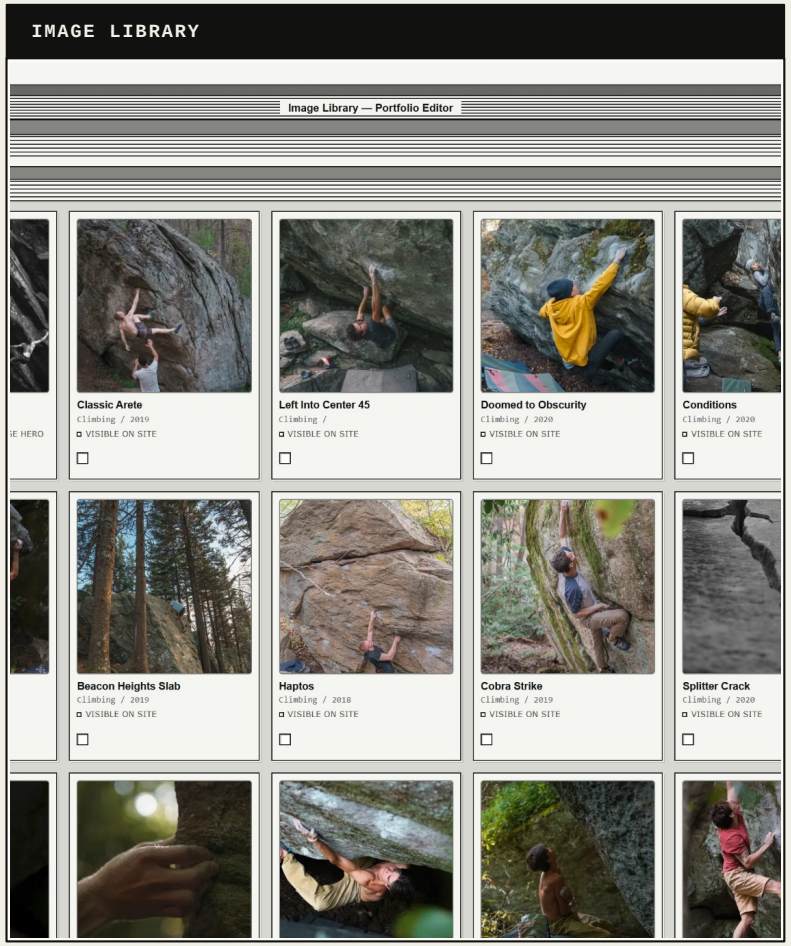
CURRENT

A data-driven website

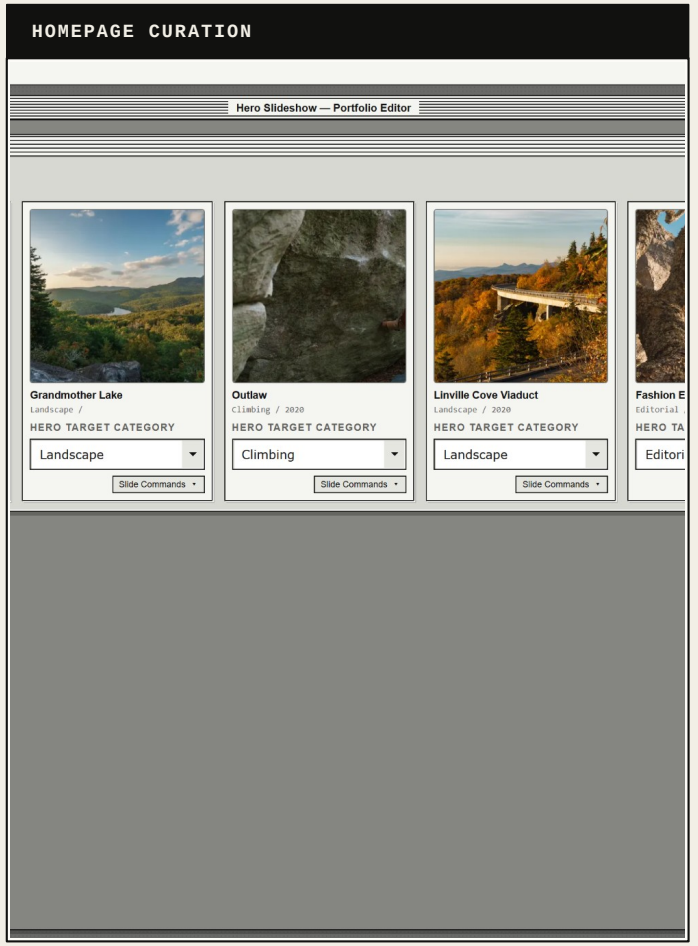
The editor changes the data the website consumes rather than rewriting its visual implementation. That keeps responsive behavior, animation, accessibility, and application logic under developer control while exposing the content that needs regular maintenance.

Images	About photography
Categories	About copy
Hero slides	SEO
Gallery curation + architecture	Entry + homepage copy

IMAGE LIBRARY



HOMEPAGE CURATION

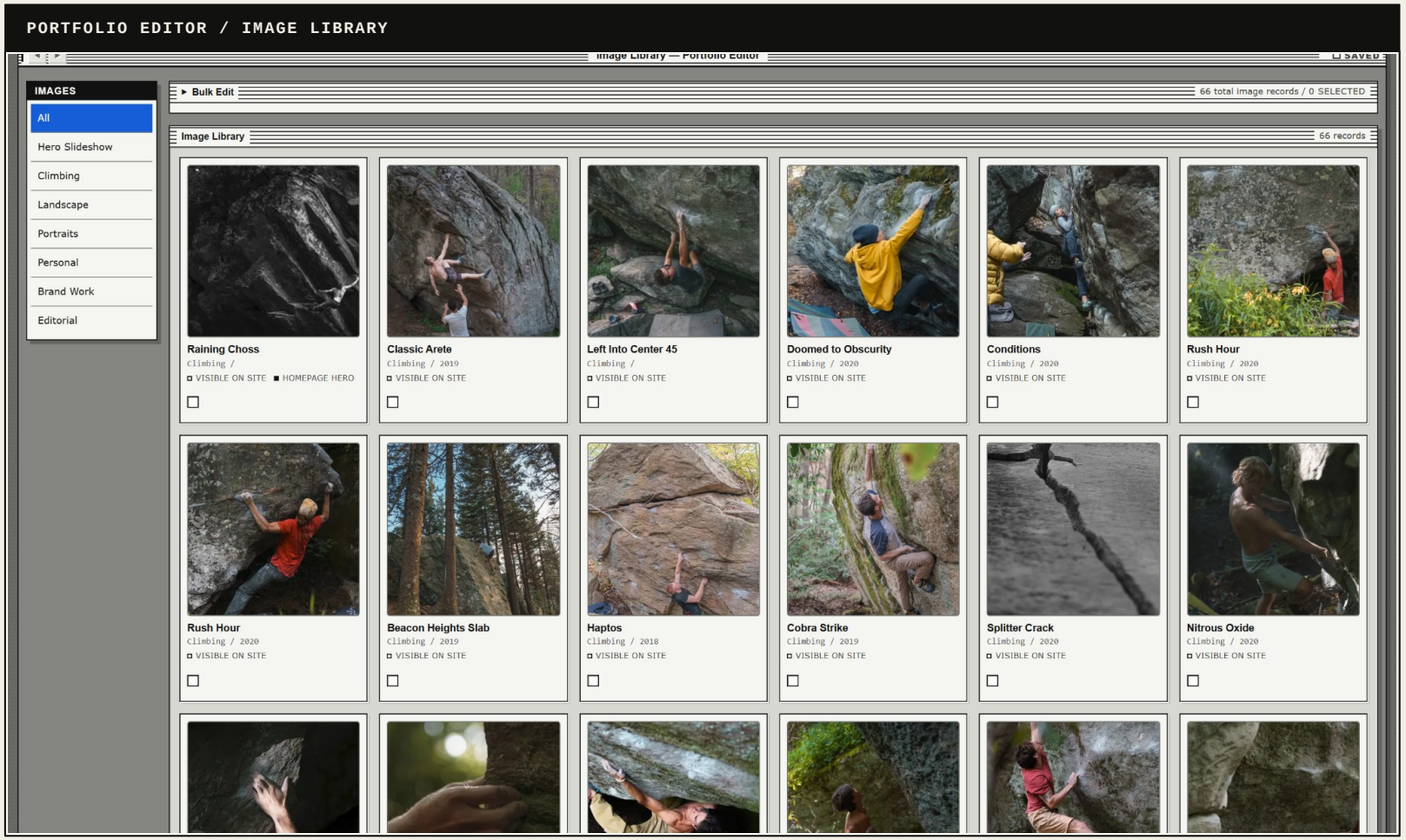


One source of truth can feed multiple presentations.

06 / EDITOR

CURRENT

The editor as a control panel



A working tool, not a static product mockup.

CONTROL MODEL

The editor combines structured controls with purpose-built visual workspaces.

Forms handle titles, metadata, categories, copy, labels, SEO, and site-wide text. Visual workspaces handle crops, slideshow ordering, collage composition, and gallery placement.

The result is not unrestricted design software. It is a focused operating layer for one highly customized site.

MEDIA	central library
DATA	structured fields
CURATION	site assignments
RECOVERY	save + backups

07 / COMPOSITION

WIP

Bounded visual editing

Not every visual decision should become unrestricted design control. The About editor exposes only the controls that matter for this page: position, size, layer, rotation, opacity, and crop.

Desktop remains the primary authored composition, tablet is treated as a preview-only breakpoint, and mobile can carry an independent arrangement when the desktop placement does not translate cleanly. The page structure, responsive logic, accessibility, and larger visual system remain intact.

SAME CONTENT

VIEW 01

Desktop arrangement

VIEW 02

Tablet preview + Mobile arrangement

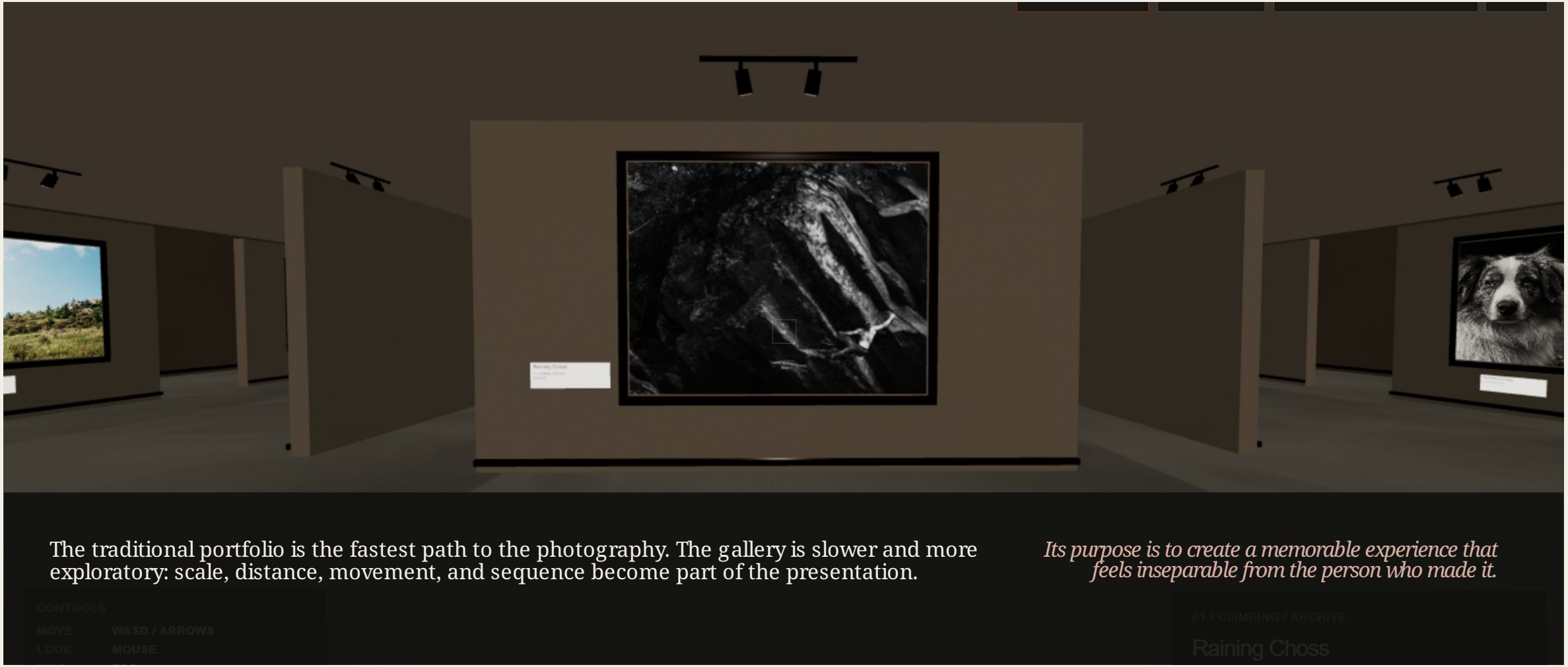
Position · size · layer · rotation · opacity

A specialized workspace can remain narrow without becoming generic design software.

08 / EXPERIENCE

WIP

Why the virtual gallery exists



The traditional portfolio is the fastest path to the photography. The gallery is slower and more exploratory: scale, distance, movement, and sequence become part of the presentation.

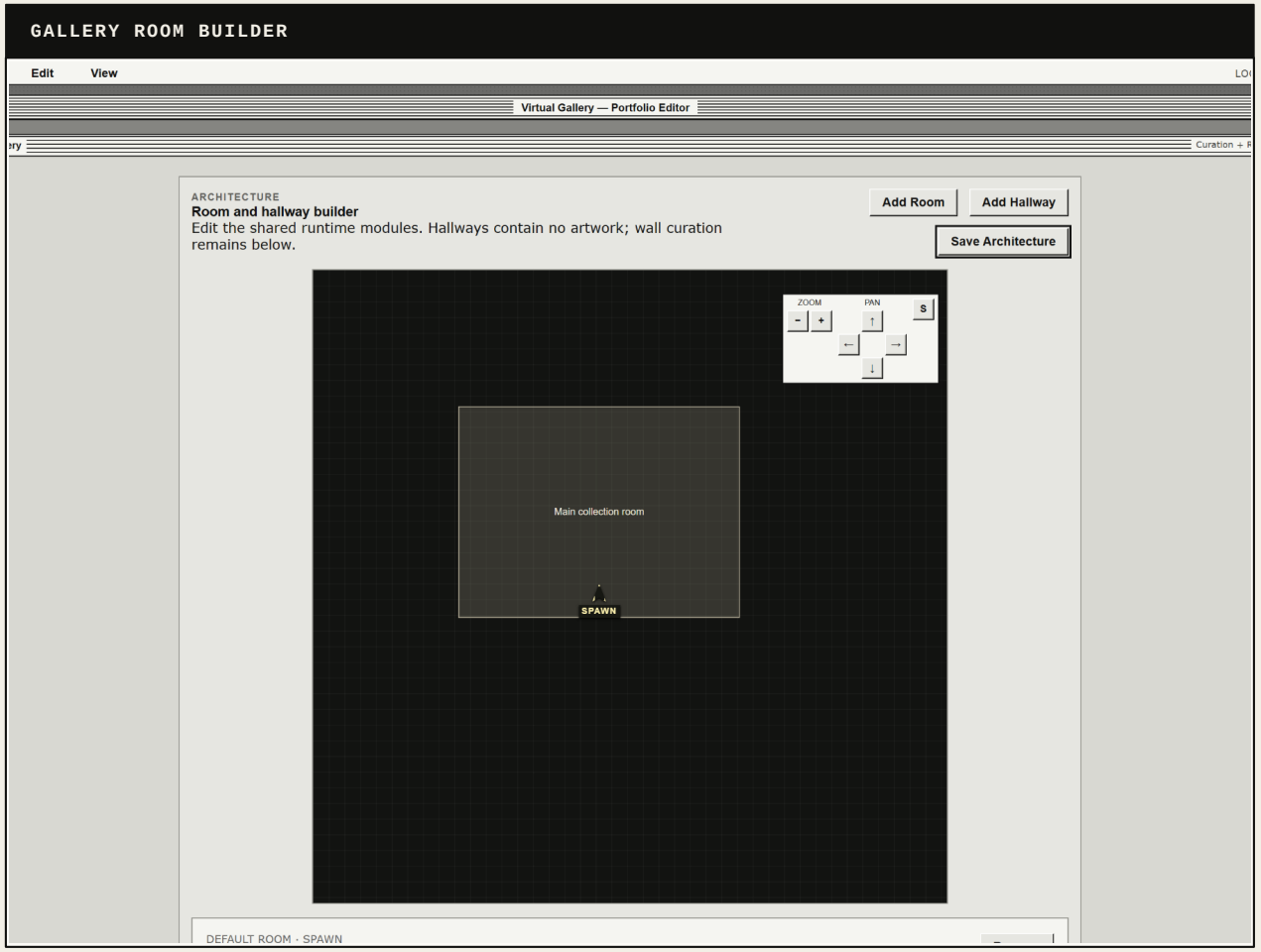
Its purpose is to create a memorable experience that feels inseparable from the person who made it.

Current gallery after Auto promoted from its Low starting state to Medium.

09 / ARCHITECTURE

WIP

Designed for expansion



Expansion was a core requirement. I did not want the first room to become a fixed environment that would need to be rebuilt when the portfolio grew.

Rooms, hallways, walls, connections, and artwork assignments are separated into editable records. The current gallery is small, but the architecture is intended to support larger exhibitions over time.

- local walls
- artwork assignments
- connection points

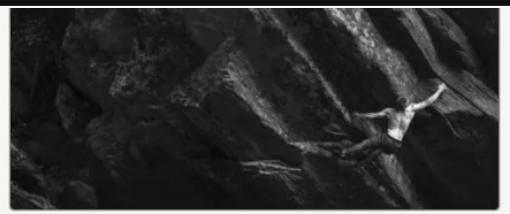
- local walls
- artwork assignments
- connection points

10 / CURATION

WIP

Specialized spatial curation

GALLERY EDITOR / WALL CURATION



MAP POSITION

Grid 0, 15 / 0.00m, 7.50m / 0°

13 × 1 blocks / bounding area 13 × 1 grid cells / 6.5m × 0.5m

HERO-SCALE WALL PREVIEW **Raining Choss** Climbing / raining-choss

6.25m × 3.60m wall / 3.26m × 2.52m frame

WALL SETTINGS

Room behavior

HERO-SCALE WALL BLOCK FOR THE STRONGEST FIRST-READ OR ANCHOR IMAGE. USES THE LARGEST WALL AND ARTWORK SCALE.

WALL BLOCK TYPE: Feature wall | DISPLAY STATUS: Active / visible | PLAQUE SIDE: Auto | ☑ Show plaque

FEATURE WALL

Hero-scale wall block for the strongest first-read or anchor image. Uses the largest wall and artwork scale.

▸ PRECISE ARTWORK ID FALLBACK

Save Wall | Move Top | Move Up | Move Down | Remove Wall

ARCHITECTURE DATA

Rooms · hallways · dimensions · connections · collision

CURATION DATA

Artwork · wall · position · scale · plaque · sequence

The workspace is an example of how a custom control panel can include a site-specific tool.

11 / PERFORMANCE

CURRENT

Adaptive quality preserves the experience

Testing on additional computers and mobile devices made the problem clear: the gallery needed to preserve its spatial experience while changing its rendering cost. Auto always begins at true Low. After sustained healthy performance and texture readiness, it may promote to Medium; a failed promotion locks it back to Low. High remains a deliberate manual tier.

CORE EXPERIENCE / PRESERVED

Room architecture
Artwork selection
Wall placement
Movement + navigation
Basic visibility

VISITOR-FACING QUALITY CONTROL

RENDERING COST / ADAPTED

Artwork texture size
Architectural lighting
ture visibility
ll-wash behavior
el ratio + anisotropy
h-only shadows

AUTO	starts Low may reach Medium
LOW	static lighting fixtures hidden
MEDIUM	pooled wall washes fixtures visible
HIGH	manual full treatment textures + shadows

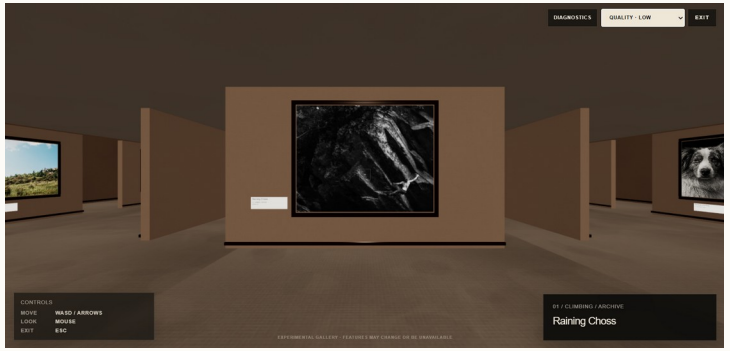
12 / EVIDENCE

CURRENT

Three tiers, one architecture

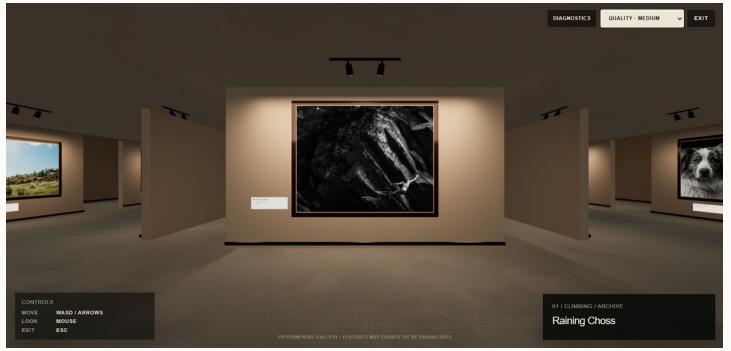
Low, Medium, and High keep the same room geometry, artwork, and navigation. The visible differences come from lighting, fixture presence, texture treatment, resolution, and shadows.

LOW



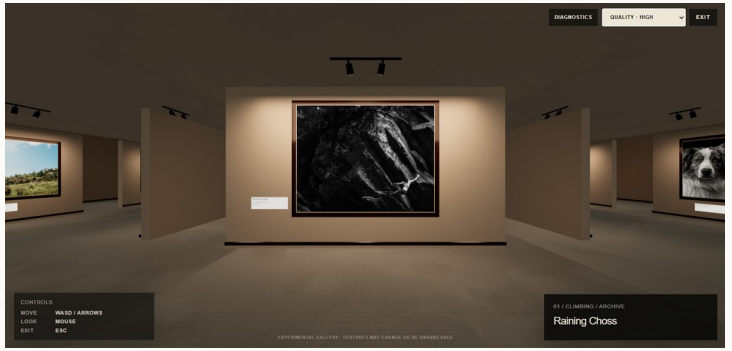
Static architectural lighting
Track fixtures hidden
0.95 pixel-ratio cap
No shadows

MEDIUM



Track fixtures visible
Pooled wall washes for visible or nearby artwork
520 px artwork textures
No shadows

HIGH



All 23 artwork lights
Up to 1.5 pixel ratio
1024 px artwork textures
Spotlight shadows

Auto starts at Low and may promote to Medium. Medium and High can look similar in a static frame because High spends more on texture detail, complete lighting, and shadows.

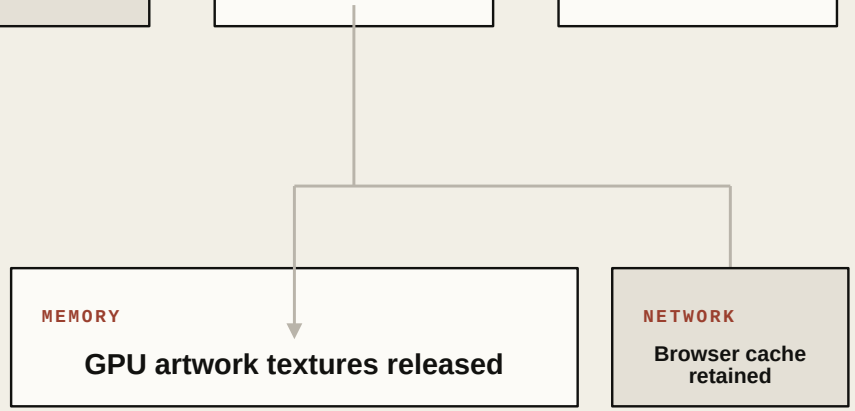
13 / DELIVERY

CURRENT

From network request to GPU resource

Browser-cache readiness and GPU readiness are separate states. Image requests can begin before gallery entry, while decode and texture upload remain controlled. On exit, artwork textures can be released from GPU memory while browser-cached files remain available for a faster return. This lifecycle supports every quality tier without permanently occupying graphics memory.

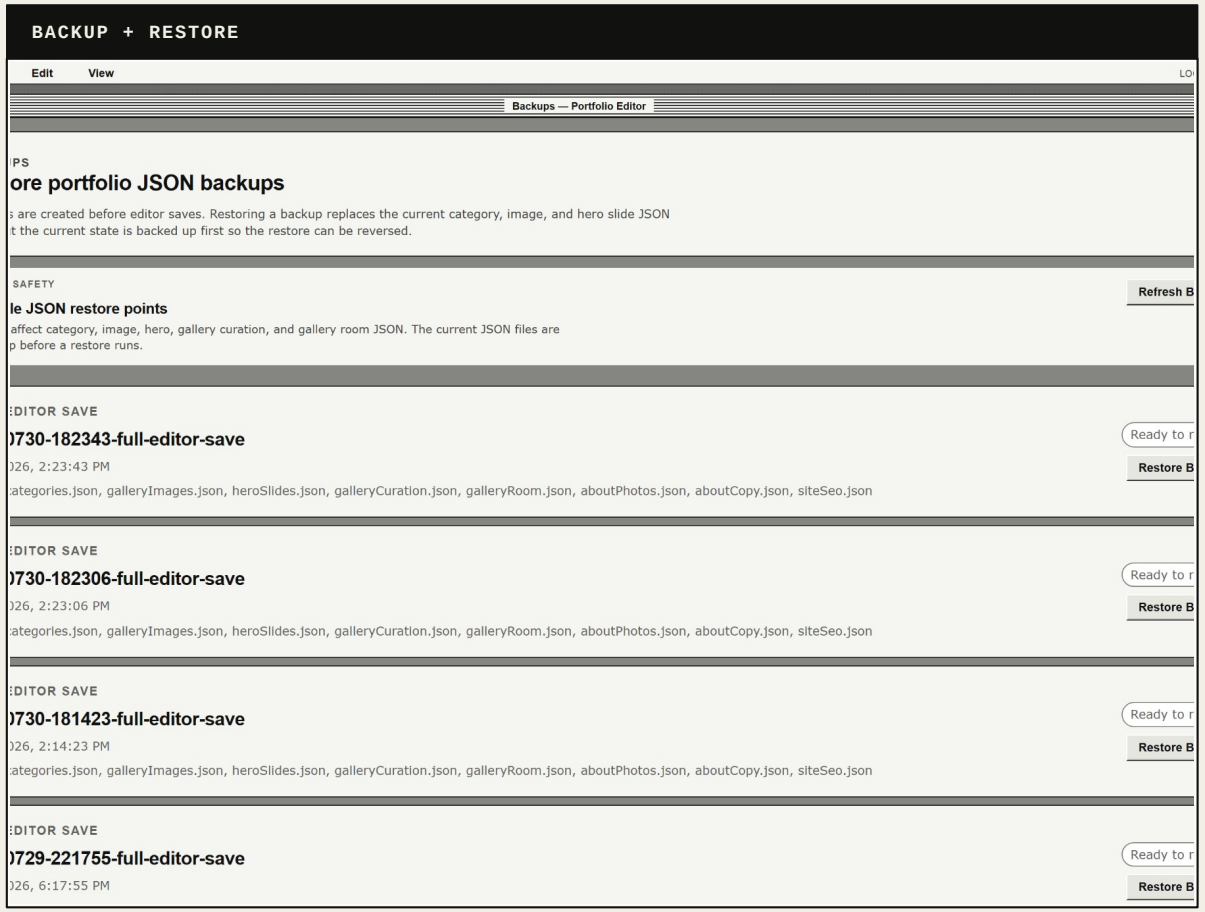
The gallery can return without paying the full network cost or permanently occupying graphics memory.



14 / RELIABILITY

WIP

Recoverable editing, honest limitations



RELIABILITY

Because the editor changes the structured data used by the public site, recoverability is essential. Save operations create backups before replacing active records. Import actions validate data and move original source files into an imported-source archive. Build, data, and layout checks help catch invalid paths and inconsistent records before release.

AI + PRODUCT DIRECTION

I used AI primarily to assist with code, debugging, and interface asset creation. I directed the project from a product-management perspective: defining problems, setting requirements, prioritizing workflows, evaluating results, and deciding how the system should behave.

The interface remains visually inconsistent and has not received a professional product-design pass. The editor is also still local, portfolio-specific, and intentionally not yet a standalone platform.

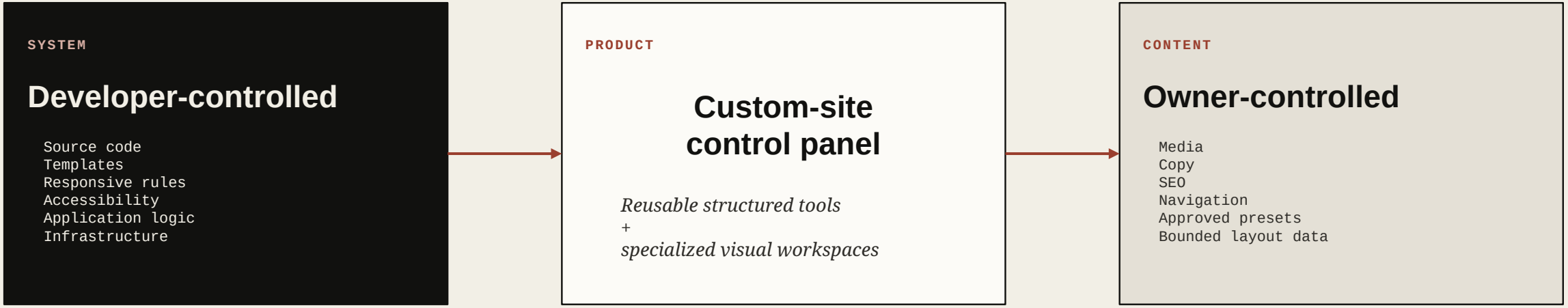
VALIDATE → BACK UP → SAVE → BUILD → REVIEW → RELEASE

15 / DIRECTION

PLANNED

From portfolio editor to custom-site control panel

The current editor was built for one website, but the product direction is broader: a schema-driven control panel that can connect to developer-built websites without handing clients unrestricted control over the implementation.



I began by building a gallery for my own photographs. The editor emerged from everything required to keep that gallery - and the website around it - usable.